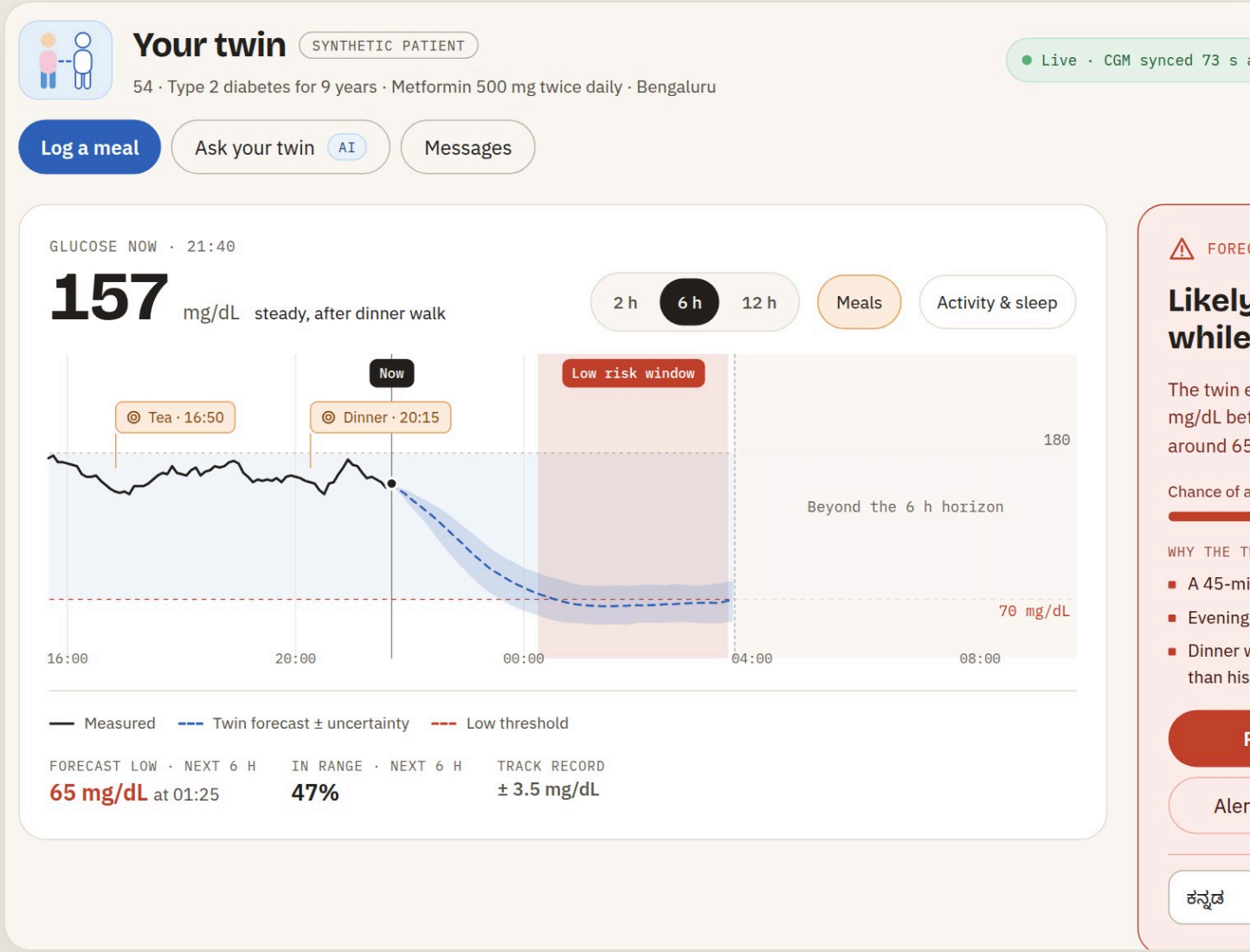
Pratibimb

A digital twin for type 2 diabetes that sees tonight's low before it happens.

A model of one person's metabolism, fitted to their own glucose, meals, walks, sleep and medicines. It looks ahead, explains why, and lets patients and doctors try a change on the twin first.



Research prototype · synthetic data · not a medical device

THE PROBLEM

Diabetes care reacts after the fact. The warning signs are already in the data.

Alarms come late

A glucose monitor alarm sounds once glucose is already out of range. Overnight lows often arrive while the person is asleep.

Advice is generic

Every body responds differently to rice, a walk, a short night's sleep or a missed dose. General rules miss what matters for this person.

Triage looks backwards

Care teams look after many patients and rank them by the last reading. Someone who looks fine now but is heading for a low tonight does not stand out.

Many of these events are predictable from the person's own patterns. Pratibimb predicts them, explains them, and helps patients and doctors act early.

OUR APPROACH

A personal digital twin: sense, mirror, simulate, act.

01
Sense
Wearables stream glucose every 5 minutes, heart rate, steps and sleep. Meals are logged by text, voice or photo.

→

02
Mirror
Health records (HbA1c, medicines) set the starting point. The twin learns this person’s physiology from 14 days of data.

→

03
Simulate
The twin plays the next 2 to 12 hours forward, with an honest uncertainty band, and replays any what-if.

→

04
Act
Nudges for the patient, such as a bedtime snack. Early alerts, reasons and drafted messages for the care team.

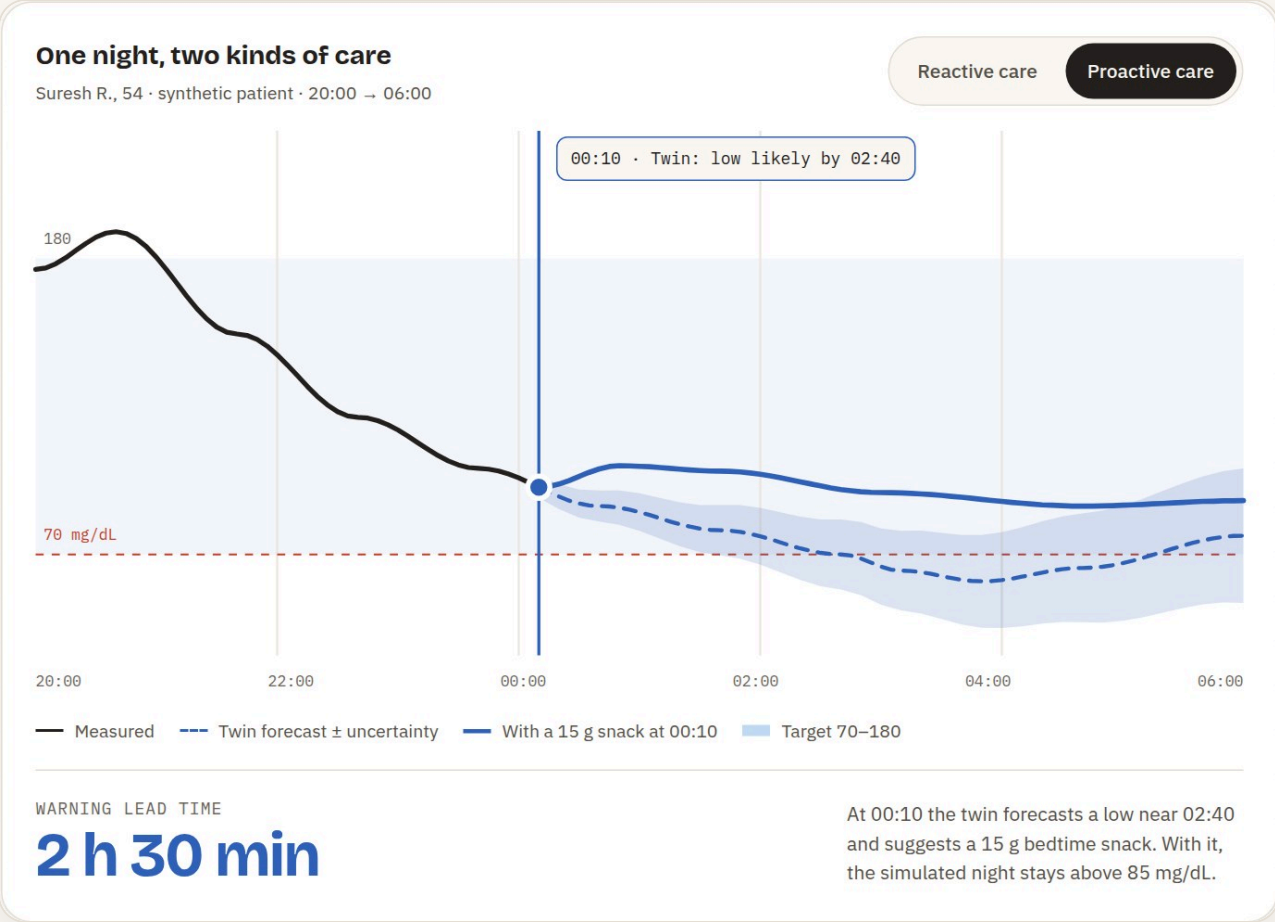
Explainable by default
Every forecast lists the signals that moved it, so a doctor can see the reasoning and disagree with it.

Consent, stream by stream
Patients switch each data source on or off and see what it is used for. Designed around India’s DPDP Act, 2023.

A clinician signs off
The twin suggests; it never changes a prescription. Any dose change goes to the patient’s doctor.

HEALTHCARE USE CASE

One night, two kinds of care.



Suresh, 54, type 2 diabetes for 9 years (synthetic). A light dinner, then a long walk.

Reactive care, today

The CGM alarm sounds at 02:40, once glucose has reached 70 mg/dL. He is asleep, and the low deepens to 60 by 03:50.

Proactive care, with the twin

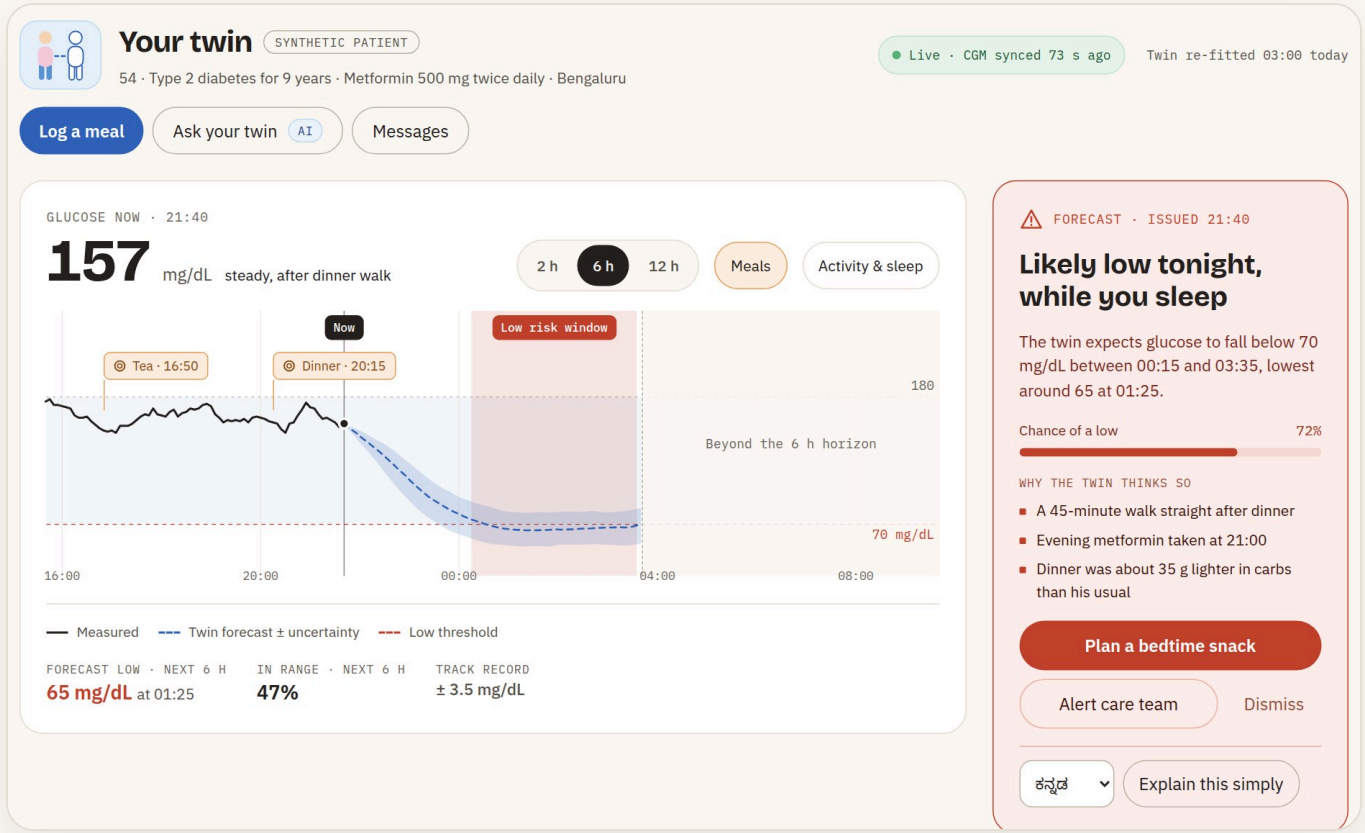
At 00:10 the twin forecasts a low near 02:40 and suggests a 15 g bedtime snack. With it, the night stays above 85 mg/dL.

WARNING LEAD TIME

2 h 30 min

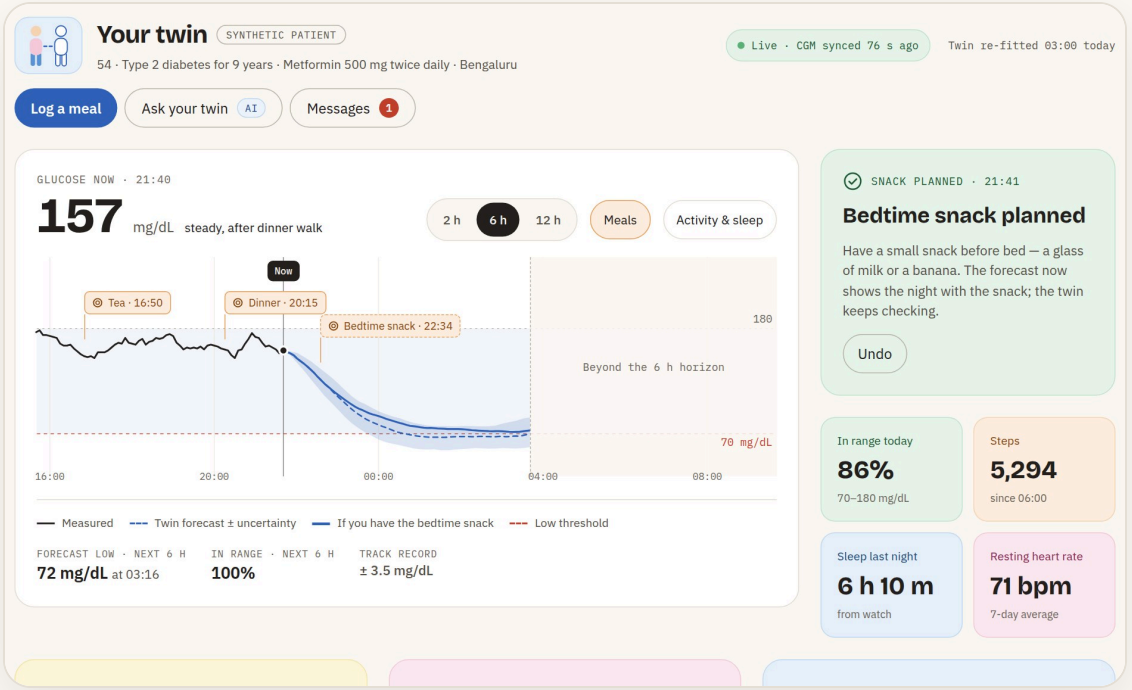
FOR THE PATIENT · LIVE TWIN

A forecast you can act on, with the reasons.

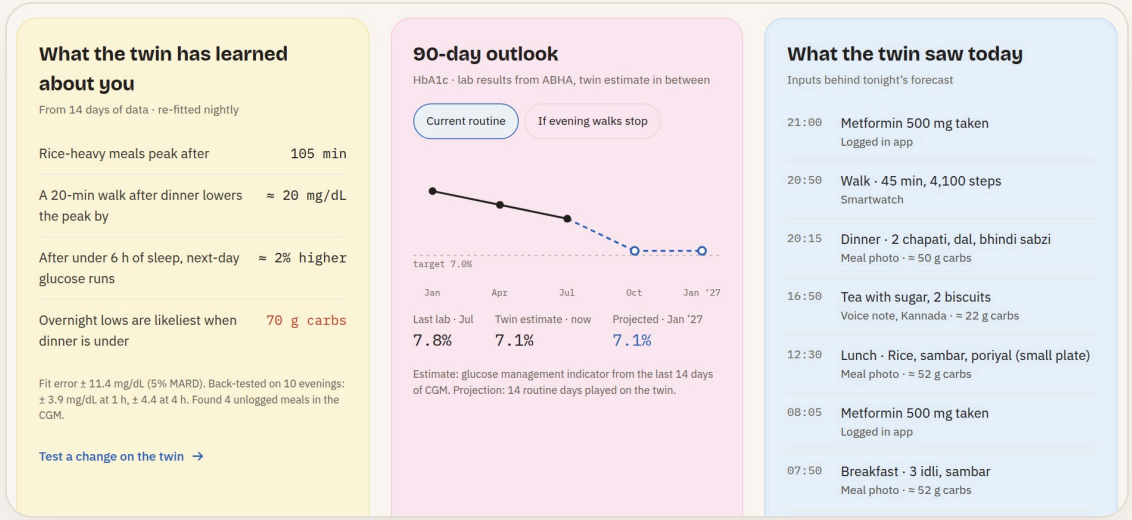


- **Measured vs forecast.** The last 6 hours from the sensor, then the twin's forecast for 2, 6 or 12 hours with an uncertainty band.
- **When and how likely.** "Likely low tonight, while you sleep": the risk window, the lowest value and the chance of a low.
- **Why the twin thinks so.** Reasons measured by replaying the evening with one thing changed, such as no walk.
- **One-tap actions.** Plan a bedtime snack, alert the care team, or dismiss. Live: new readings every 5 minutes.

See the night redrawn. See what the twin learned about you.



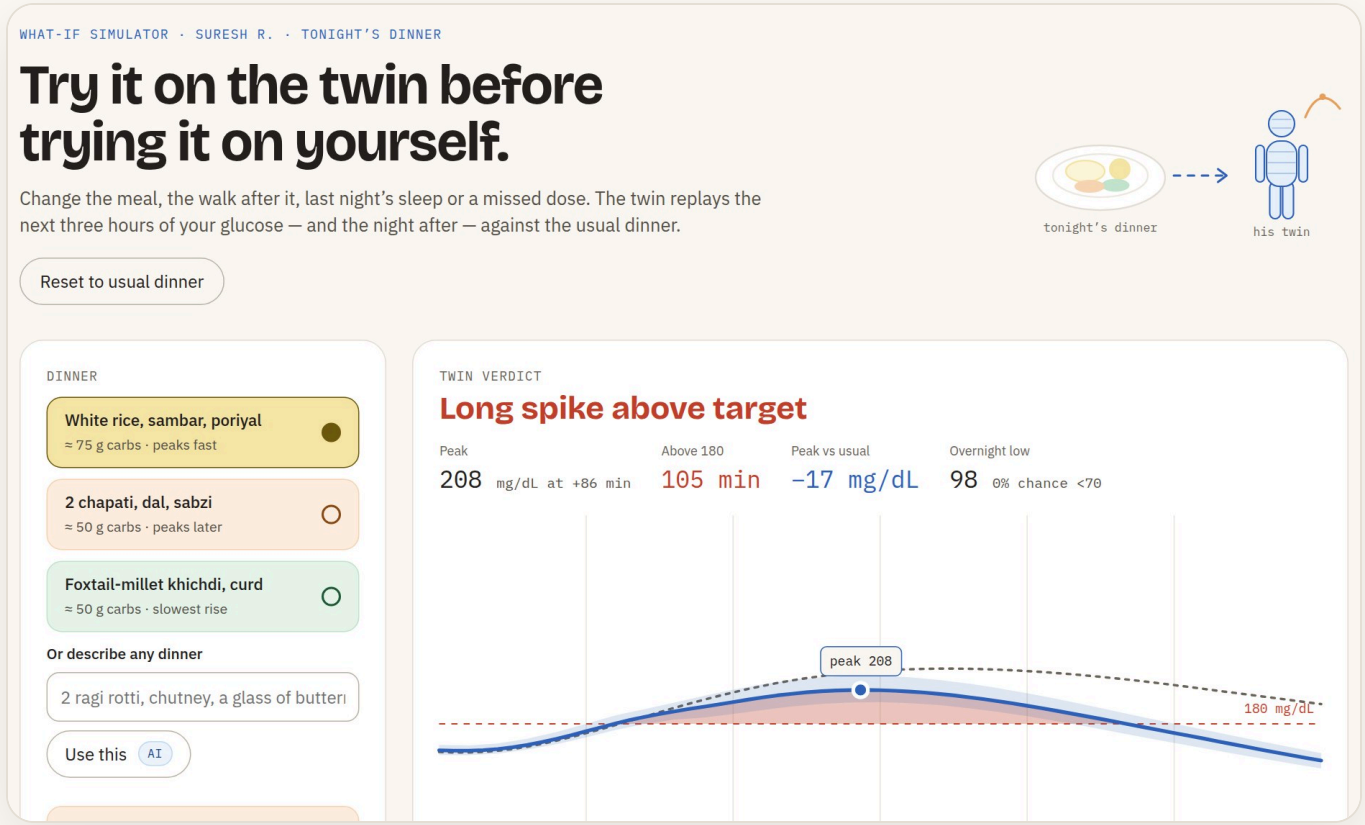
Bedtime snack planned. A second line shows the night with the snack; the twin keeps checking and can be undone.



Personal, not generic. Meal peaks, the walk effect, the sleep effect, an HbA1c outlook, and meals the twin found in the CGM that were never logged.

FOR THE PATIENT · WHAT-IF

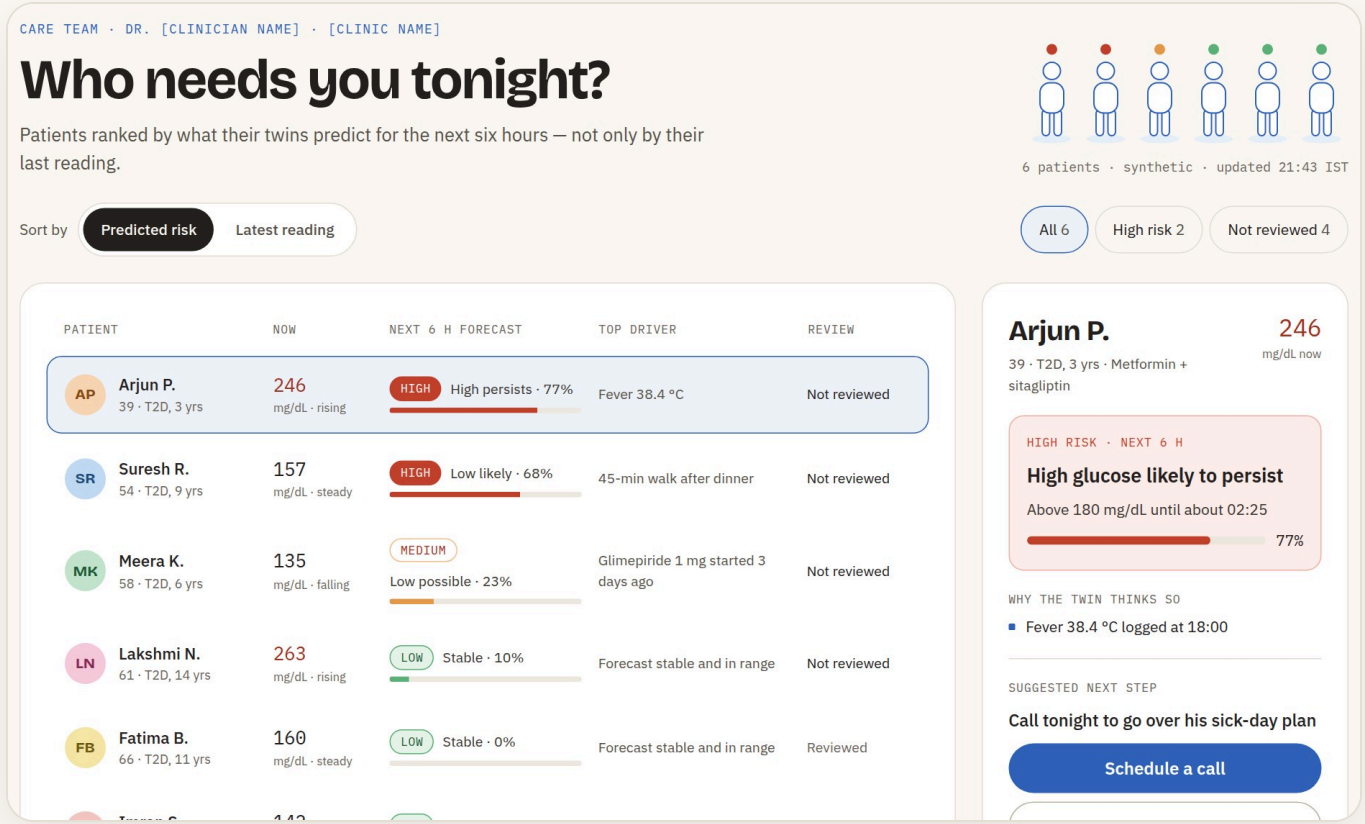
Try it on the twin before trying it on yourself.



- **Change one thing.** Dinner (rice, chapati, millet or any dinner described in words), portion, the walk after it, last night's sleep, or a missed evening dose.
- **A clear verdict.** Peak, minutes above 180, change vs the usual dinner, overnight low and its chance.
- **Compare.** Save up to four versions of tonight side by side.
- **Dinner ideas ranked by the twin.** AI suggests home-style dinners; the twin plays each one and ranks them.

FOR THE CARE TEAM

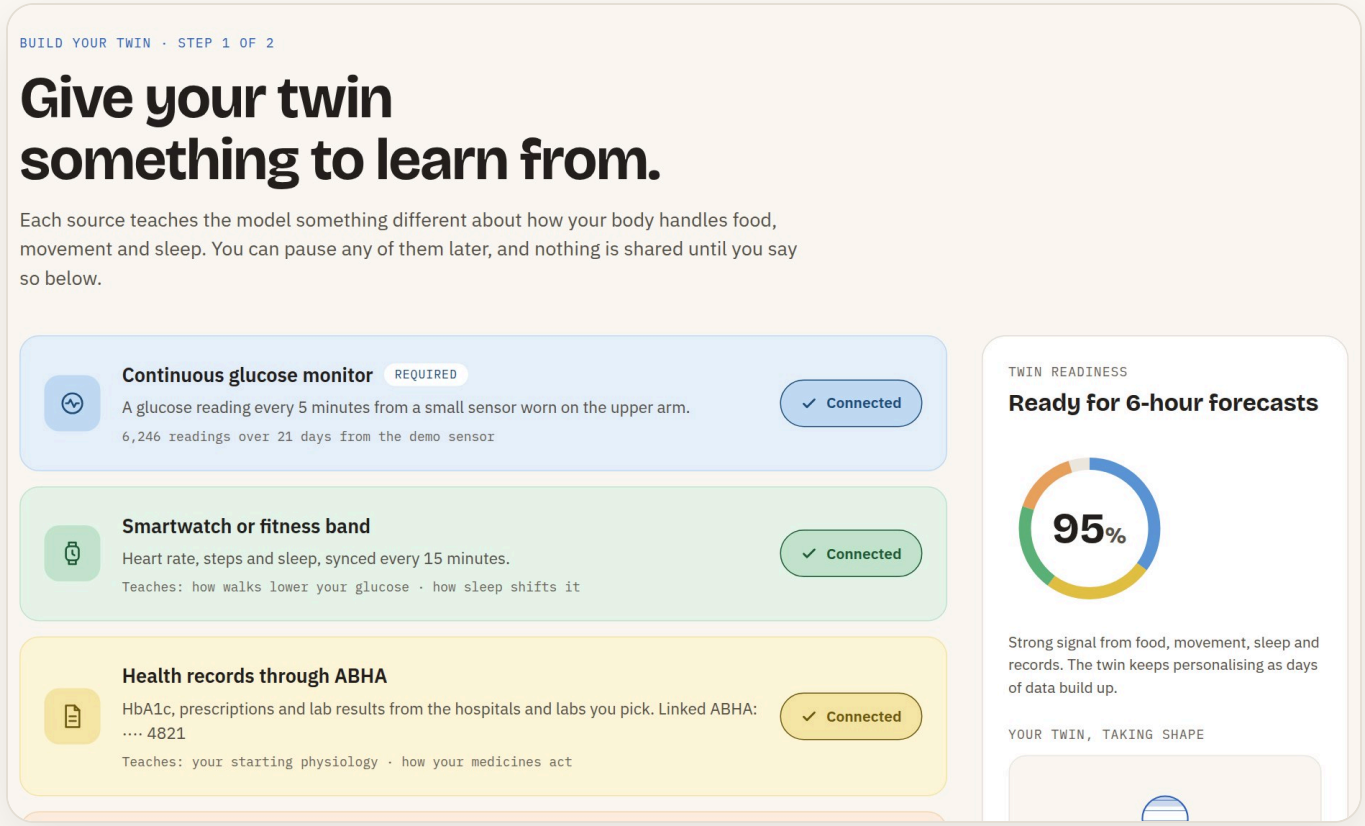
Who needs you tonight? Ranked by prediction, not the last reading.



- **Predicted risk first.** Suresh's 157 mg/dL looks fine, yet he ranks High: his twin expects a low tonight.
- **Reasons and a next step.** Arjun has a fever; his twin expects the high to persist, and suggests a call about his sick-day plan.
- **Assistant.** A handoff note in SBAR format, and a message drafted in the patient's language for the doctor to edit and send.
- **Review workflow.** Filters for high risk and not reviewed; any patient's full twin one click away.

ONBOARDING AND CONSENT

A new patient has a working twin in minutes.



- **Five sources, each switched on separately:** glucose monitor, smartwatch, ABHA health records, meal log, blood-pressure cuff.
- **Real exports work.** Import a Dexcom Clarity or LibreView CSV, or use the demo sensor.
- **Records read by AI.** A lab report or prescription becomes structured results that the patient reviews before saving.
- **Build my twin.** Fitted in seconds, then re-fitted every night at 03:00.

GENERATIVE AI · GOOGLE GEMINI

AI that talks in the patient's language, and only quotes the twin.

Meal logging

From a photo, or a description typed or spoken in English, हिन्दी or ಕನ್ನಡ. Items and carbs, checked before saving.

Ask your twin

A chat agent with five tools that run this patient's forecast, what-ifs and learned facts before it answers.

Explain simply

Tonight's forecast as a short note in the patient's language.

Records to data

Lab reports and prescriptions (PDF or photo) to structured labs and medicines.

Dinner ideas

The model suggests dinners; the twin plays and ranks them.

Care-team assistant

SBAR handoff notes and drafted messages that the clinician edits and sends.

Guardrails: never suggests a dose change · every number comes from the twin · structured output checked against a schema · blocked answers never shown as real · offline fallbacks (Indian food table, templates) when the AI is unavailable.

A physiological model, personalised and honest about its error.

Physiology

Bergman-style minimal model with a pancreas, extended with two-speed meal absorption, walk effects, metformin and sulfonylurea pharmacokinetics, sleep debt, illness and the dawn rise.

Personal fit

10 parameters per person from 14 days of CGM. Maximum a posteriori estimate with a population prior, optimised with Nelder–Mead; Laplace approximation for uncertainty.

Unlogged meals

Rises no logged meal explains are detected and sized by least squares, then the model is refitted (EM-style). The patient confirms or rejects them.

Probabilistic forecast

Hidden state estimated now, then 60 plausible versions of the person played forward with day-to-day variability: median, band, P(low), P(high), risk window.

Self-calibration

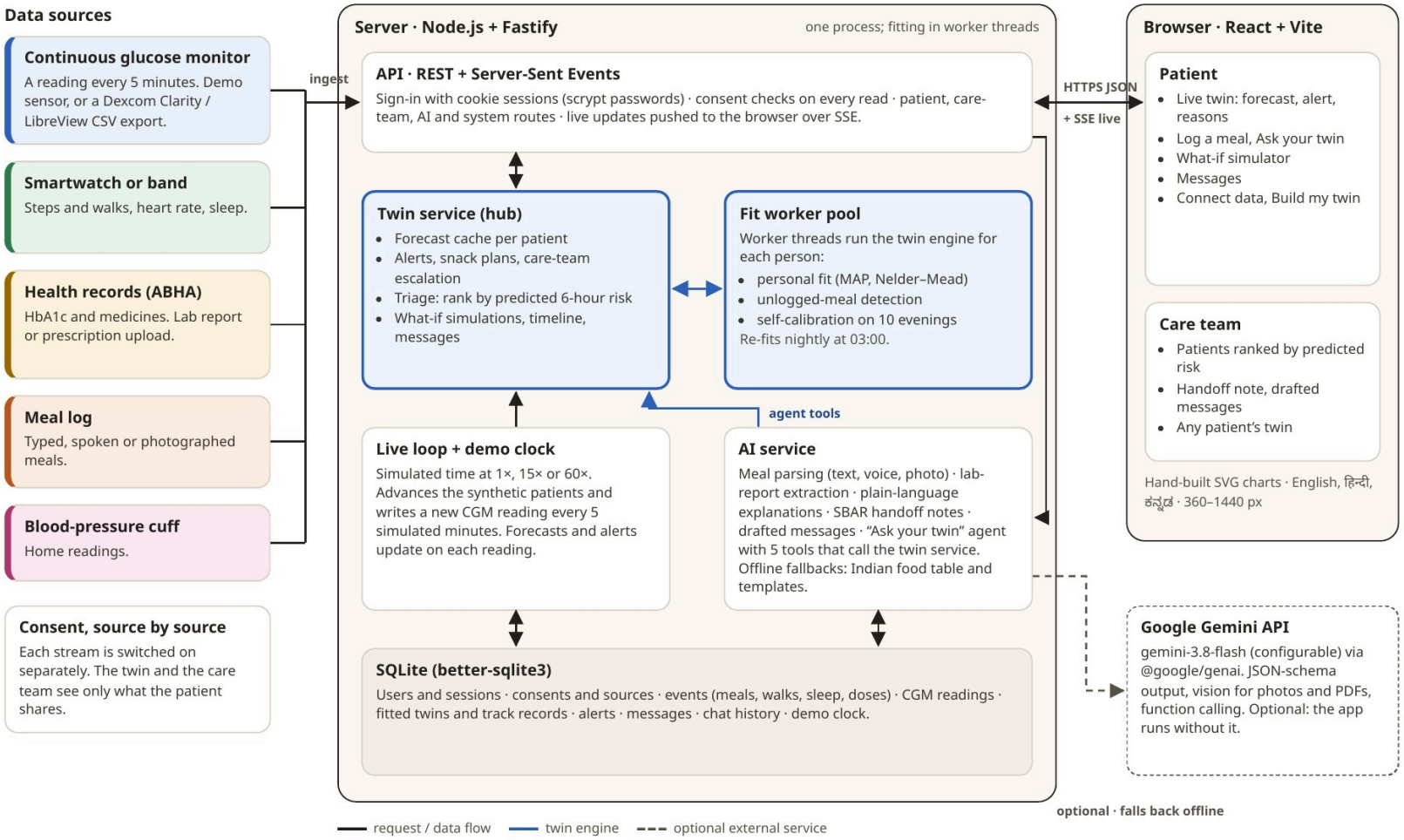
After each fit, the twin back-tests itself on its last 10 evenings and widens or shifts its forecasts by its own measured error.

Explanations

Counterfactual replays measure how much each factor (dinner size, the walk, a missed dose) moved tonight’s forecast.

PRATIBIMB · ARCHITECTURE 1 OF 2 From a patient's devices to a forecast on their phone and a ranked list for the care team.

System overview



Pratibimb · type 2 diabetes digital twin

Research prototype · synthetic data · not a medical device

TECHNICAL STACK

Lightweight, open and easy to run.

LAYER	TECHNOLOGY
Frontend	React 18, TypeScript, Vite, React Router · hand-built SVG charts · design tokens to CSS variables
Backend	Node.js, Fastify, Server-Sent Events for live updates, cookie sessions with scrypt password hashing
Database	SQLite (better-sqlite3)
Twin engine	Plain TypeScript shared by server, browser and tests; fitting in Node worker threads
Generative AI	Google Gemini (gemini-3.8-flash, configurable) via @google/genai: JSON-schema output, vision, function calling
Quality	zod validation, Vitest (128 tests), ESLint with accessibility rules
Deployment	One Node process serves API and app; Docker and docker-compose; Windows launcher

OUTCOMES

What we built, and how well the twin performs.

TWIN ACCURACY (6 SYNTHETIC PATIENTS, 14 DAYS EACH)	RESULT
Fit error (RMSE)	**6.6–12.4 mg/dL**
Mean absolute relative difference	**3.6–5.3%**
Back-tested forecast error, 1 h ahead	**2.9–4.8 mg/dL**
Back-tested forecast error, 4 h ahead	**3.0–6.8 mg/dL**
Unlogged meals found in the CGM	**2–5 per patient**
Overnight low flagged ahead of the risk window	**2 h 35 min**

Back-tests use 9–10 evenings per patient. The synthetic patients and the twin share a model family, so these figures are optimistic; validation on real CGM data is the next step.

A complete working product

Patient app, care-team view, onboarding with consent, live demo clock, six realistic synthetic patients.

Triage that looks ahead

Suresh (overnight low) and Arjun (fever, persistent high) ranked High; one Medium; three stable patients Low.

Built for everyone

128 automated tests · English, Hindi, Kannada · keyboard and screen-reader support · 44 px targets · 4.5:1 contrast · 360 to 1440 px.

SAFETY, PRIVACY AND TRUST

Built to be questioned.

The doctor decides

The twin suggests; it never changes a prescription. The AI never suggests a dose change. Messages are drafted, then edited and sent by the clinician.

Patients own their data

Consent per source, visible use of each stream, designed around the DPDP Act, 2023. The care team sees only what the patient shares.

Numbers you can trace

Every number shown or quoted by the AI comes from the twin. Forecasts carry an uncertainty band and a published track record.

Works without AI

Forecasts, what-ifs and triage run locally. If the AI service is off or busy, the app falls back to a food table and templates.

WHAT COMES NEXT

From prototype to pilot.

Step 1

Validate

Measure forecast error on real CGM data and compare with the synthetic results.

→

Step 2

Integrate

Real CGM and wearable feeds, ABHA / ABDM health records, writing to the same tables the demo uses.

→

Step 3

Pilot

A care-team pilot, with Hindi and Kannada messages reviewed by native speakers.

→

Step 4

Extend

Hypertension (morning blood-pressure surges) and cardiovascular risk on the same twin approach.

THANK YOU

Pratibimb

Early warnings instead of late alarms, in the patient's own language, with the doctor in charge.

Demo video	youtube.com/watch?v=veTwt_dNIos
Licence	MIT, open source

Research prototype · all patients shown are synthetic · not a medical device